

DL Assignment 10

Name - KAVISH Paraswar

Class - CS(AI) B

Batch - 01

Roll no - 2

PRN - 12311694

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```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("sumanthvrao/daily-climate-time-series-data")

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/sumanthvrao/daily-climate-
100%|██████████| 22.0k/22.0k [00:00<00:00, 30.1MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/sumanthvrao/daily-
```

```
import pandas as pd
import os

path = "/root/.cache/kagglehub/datasets/sumanthvrao/daily-climate-time-series-c
print(os.listdir(path))
```

```
['DailyDelhiClimateTrain.csv', 'DailyDelhiClimateTest.csv']
```

```
df = pd.read_csv(os.path.join(path, "DailyDelhiClimateTrain.csv"))

df.head()
```

	date	meantemp	humidity	wind_speed	meanpressure
0	2013-01-01	10.000000	84.500000	0.000000	1015.666667
1	2013-01-02	7.400000	92.000000	2.980000	1017.800000
2	2013-01-03	7.166667	87.000000	4.633333	1018.666667
3	2013-01-04	8.666667	71.333333	1.233333	1017.166667
4	2013-01-05	6.000000	86.833333	3.700000	1016.500000

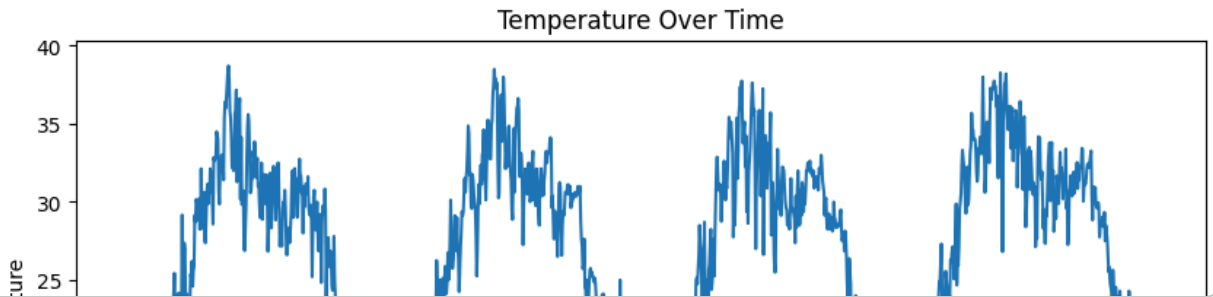
```
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1462 entries, 0 to 1461
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   date             1462 non-null   object
1   meantemp         1462 non-null   float64
2   humidity         1462 non-null   float64
3   wind_speed       1462 non-null   float64
4   meanpressure     1462 non-null   float64
dtypes: float64(4), object(1)
memory usage: 57.2+ KB
None
```

	meantemp	humidity	wind_speed	meanpressure
count	1462.000000	1462.000000	1462.000000	1462.000000
mean	25.495521	60.771702	6.802209	1011.104548
std	7.348103	16.769652	4.561602	180.231668
min	6.000000	13.428571	0.000000	-3.041667
25%	18.857143	50.375000	3.475000	1001.580357
50%	27.714286	62.625000	6.221667	1008.563492
75%	31.305804	72.218750	9.238235	1014.944901
max	38.714286	100.000000	42.220000	7679.333333

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))
plt.plot(df['meantemp'])
plt.title("Temperature Over Time")
plt.xlabel("Days")
plt.ylabel("Temperature")
plt.show()
```



```
from sklearn.preprocessing import MinMaxScaler
import numpy as np

data = df[['meantemp']].values

scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(data)
```

```
X = []
y = []

time_step = 30

for i in range(time_step, len(scaled_data)):
    X.append(scaled_data[i-time_step:i, 0])
    y.append(scaled_data[i, 0])

X, y = np.array(X), np.array(y)
```

```
X = X.reshape(X.shape[0], X.shape[1], 1)

print("Shape of X:", X.shape)
```

```
Shape of X: (1432, 30, 1)
```

```
split = int(0.8 * len(X))

X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, LSTM

model = Sequential()

model.add(LSTM(50, return_sequences=True, input_shape=(X.shape[1],1)))
model.add(LSTM(50))
model.add(Dense(1))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199:
    super().__init__(**kwargs)
```

```
model.compile(  
    optimizer='adam',  
    loss='mean_squared_error'  
)
```

```
model.fit(  
    X_train, y_train,  
    epochs=20,  
    batch_size=32  
)
```

```
Epoch 1/20  
36/36  5s 24ms/step - loss: 0.0546  
Epoch 2/20  
36/36  2s 38ms/step - loss: 0.0074  
Epoch 3/20  
36/36  2s 46ms/step - loss: 0.0058  
Epoch 4/20  
36/36  2s 56ms/step - loss: 0.0055  
Epoch 5/20  
36/36  2s 41ms/step - loss: 0.0054  
Epoch 6/20  
36/36  1s 26ms/step - loss: 0.0053  
Epoch 7/20  
36/36  1s 26ms/step - loss: 0.0052  
Epoch 8/20  
36/36  1s 28ms/step - loss: 0.0052  
Epoch 9/20  
36/36  1s 26ms/step - loss: 0.0052  
Epoch 10/20  
36/36  1s 26ms/step - loss: 0.0047  
Epoch 11/20  
36/36  1s 38ms/step - loss: 0.0047  
Epoch 12/20  
36/36  3s 84ms/step - loss: 0.0046  
Epoch 13/20  
36/36  2s 53ms/step - loss: 0.0047  
Epoch 14/20  
36/36  2s 24ms/step - loss: 0.0047  
Epoch 15/20  
36/36  1s 24ms/step - loss: 0.0044  
Epoch 16/20  
36/36  1s 24ms/step - loss: 0.0043  
Epoch 17/20  
36/36  1s 24ms/step - loss: 0.0043  
Epoch 18/20  
36/36  1s 25ms/step - loss: 0.0043  
Epoch 19/20  
36/36  3s 78ms/step - loss: 0.0041  
Epoch 20/20  
36/36  5s 126ms/step - loss: 0.0041  
<keras.src.callbacks.history.History at 0x795d8e1d11c0>
```

```
predictions = model.predict(X_test)
```

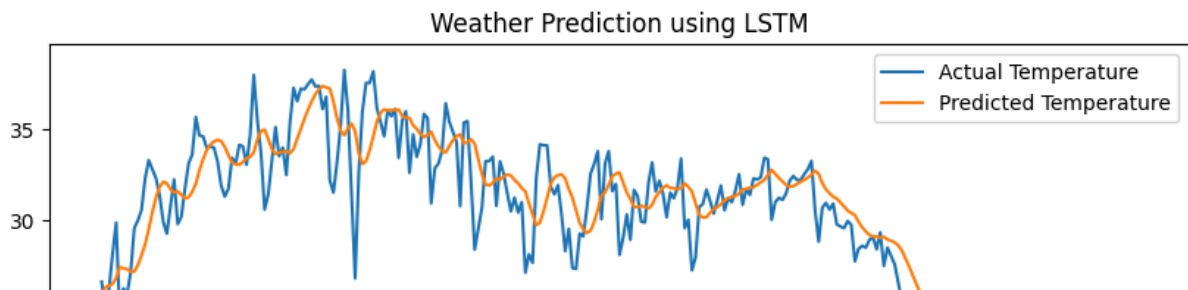
```
# Convert back to original scale
predictions = scaler.inverse_transform(predictions)
y_test_actual = scaler.inverse_transform(y_test.reshape(-1,1))
```

9/9 ————— 1s 66ms/step

```
plt.figure(figsize=(10,5))

plt.plot(y_test_actual, label='Actual Temperature')
plt.plot(predictions, label='Predicted Temperature')

plt.legend()
plt.title("Weather Prediction using LSTM")
plt.show()
```



```
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(y_test_actual, predictions)
rmse = np.sqrt(mse)

print("RMSE:", rmse)
```

RMSE: 2.0202752054351136